

SMART APS

Advanced Planning and Scheduling System

Constraints and Solutions Document

Document Title	Smart APS — Constraints and Solutions
Organisation	Uno Minda Rika Private Limited
Department	Injection Moulding Division
Prepared By	Tushar (Intern)
Guided By	Mr. Amrit Lal
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This document explains every constraint built into the Smart APS system — what the problem is, how the system solves it, and what happens when the primary solution is not sufficient.

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1. INTRODUCTION AND PURPOSE

Smart APS (Advanced Planning and Scheduling System) is a software tool developed under the guidance of Mr. Amrit Lal to automate the daily production plan for the Injection Moulding Division. It runs every day, reads the current inventory levels and monthly targets from the existing Excel data files, and produces a complete, ready-to-execute production plan within sixty seconds.

Every good planning system operates within a set of rules, called constraints. These constraints exist because the real factory floor has real limitations: machines can only run for so many hours, only one changeover can happen at a time, and some parts are more urgent than others. Without these rules, a plan would be mathematically possible on paper but impossible to execute in practice.

This document explains each of the ten constraints built into Smart APS V9. For each constraint, the following three-part structure is used:

THE PROBLEM	What goes wrong without this rule.
OUR SOLUTION	What Smart APS does to prevent the problem.
IF IT STILL FAILS	The backup plan when the primary solution is not sufficient.

2. CONSTRAINT 1: PLAN QUANTITIES BASED ON INVENTORY HEALTH

The Problem

Not every day is the same. On some days, all parts have several days of stock sitting in the warehouse and the situation is comfortable. On other days, several parts are at zero stock and any delay in production could stop the supply chain. If the planning system treats both situations identically, it will either over-produce when things are healthy or under-react when there is a crisis.

Our Solution

Every day, before building the plan, Smart APS assesses the inventory health of all active parts and classifies the day into one of four situations, called scenarios. In V9, two inventory constants govern this logic: SAFETY_DAYS (3 days) is the urgency floor — a part below this is consuming its safety buffer and is treated as urgent. TARGET_DAYS (5 days) is the absolute ceiling — no part is ever produced beyond five times its daily indent. The five-day target comprises three days of safety buffer plus two days of additional cushion to cover days when a part may not be planned due to machine constraints. Parts already at or above the five-day ceiling are skipped entirely on that day. Table 2 describes the four scenarios.

Scenario	Condition	What the System Does	Quantity Target
S0 — Crisis	All parts have less than one day of stock	Produce toward the five-day full target (three-day safety buffer plus two days safety cushion). Most urgent parts assigned first. Every machine hour directed at closing the gap.	Up to 5 days of stock (3-day safety buffer + 2-day safety cushion)
S1 — Mixed	Some parts are critical; others have stock	Critical parts produced first toward the five-day target. Parts with some stock still produced to	Up to 5 days of stock for all active parts

		close their remaining gap to the five-day target.	
S2 — Recovery	All parts exist but are below the five-day full target	Produce enough to bring every part up to the five-day target. The gap is closed systematically across all machines.	Up to 5 days of stock (full five-day target)
S3 — Healthy	All parts already have five or more days of stock	Produce only one day of supply for each part to replace today's consumption. Parts already at the five-day ceiling are skipped entirely. Machines run their minimum four-hour block if daily indent is small.	Exactly 1 day of supply (daily requirement only)

Table 2: The four production scenarios and their corresponding planning logic.

If It Still Fails

If despite the scenario-adjusted planning some parts still cannot be produced due to machine capacity constraints, they appear in the VT_Not_Planned report. The current scenario is printed clearly at the top of every output sheet so the planner always understands the overall factory situation at a glance. A new output sheet, VT_Inventory_Target, shows every part against its five-day target with gap, buffer status, and estimated days to reach the target.

3. CONSTRAINT 2: MOST URGENT PARTS MUST BE PLANNED FIRST

The Problem

If parts are assigned to machines in a random or arbitrary order, a critically urgent part with zero inventory might end up being scheduled after a less important part. By the time the urgent part reaches the front of the queue, all suitable machines may already be fully occupied. The most critical part misses the day entirely, which is the worst possible outcome.

Our Solution

Before any scheduling begins, Smart APS calculates a priority score between zero and one hundred for every active part. Parts are then assigned to machines in order of their score, from highest to lowest. This guarantees that the most urgent part is always the first to receive a machine. The score is based on three factors, as shown in Table 3.

Factor	Weight in Score	How It Is Calculated
Urgency: how far below the three-day inventory target is this part?	55 percent	A part below the three-day safety floor scores the full 55 points. A part above the safety floor scores zero on this factor, even if it has not yet reached the five-day ceiling.
Category: is it a Runner, Repeater, or Stranger?	25 percent	A Runner (daily high-frequency part) scores 25 points. A Repeater scores 15. A Stranger scores 5.
Relative demand: how much of this part is needed compared to all others?	20 percent	The part with the highest daily demand scores the full 20 points. Others are scored proportionally.

Table 3: Priority score components and their weightings.

A part with zero inventory that is also a Runner will score close to 100 out of 100, and will always be among the first parts assigned to a machine, regardless of any other consideration.

In V9, if a zero-inventory Runner or Repeater still cannot find a machine after the main scheduling pass, the system runs a Displacement Pass: it identifies the compatible machine carrying the part with the most days of stock (the least urgent planned part), reduces that part's run hours by the minimum run block, and assigns the freed slot to the zero-inventory part. The displaced part must have at least three days of stock before it can be displaced — a part in its own safety buffer cannot be displaced to rescue another.

If It Still Fails

Even with the highest possible priority score, a part may not be planned if all compatible machines are genuinely full. In this case, the part appears at the top of the VT_Not_Planned report with its score clearly visible, so the planner can immediately see what is most urgent and take action.

4. CONSTRAINT 3: THE 22-HOUR DAILY PRODUCTION RULE

The Problem

The factory runs two shifts of 12 hours each every day. Each shift includes a one-hour break for tea and lunch, leaving 11 effective working hours per day. Across both shifts, the total available production time per machine per day is 22 hours. Without a strict rule, a planner could accidentally assign more work to a machine than it can physically complete in these 22 hours. The extra parts would never be made, and nobody would know until the shift was already underway.

Our Solution

Before assigning any part to a machine, Smart APS checks exactly how many hours have already been allocated to that machine across both shifts. It calculates the run time required for the part and the time needed for the mould changeover, adds them together, and only proceeds if the total fits within the remaining hours out of the 22 available. The machine can never be over-committed.

The logic the system follows is shown in Figure 1 below.

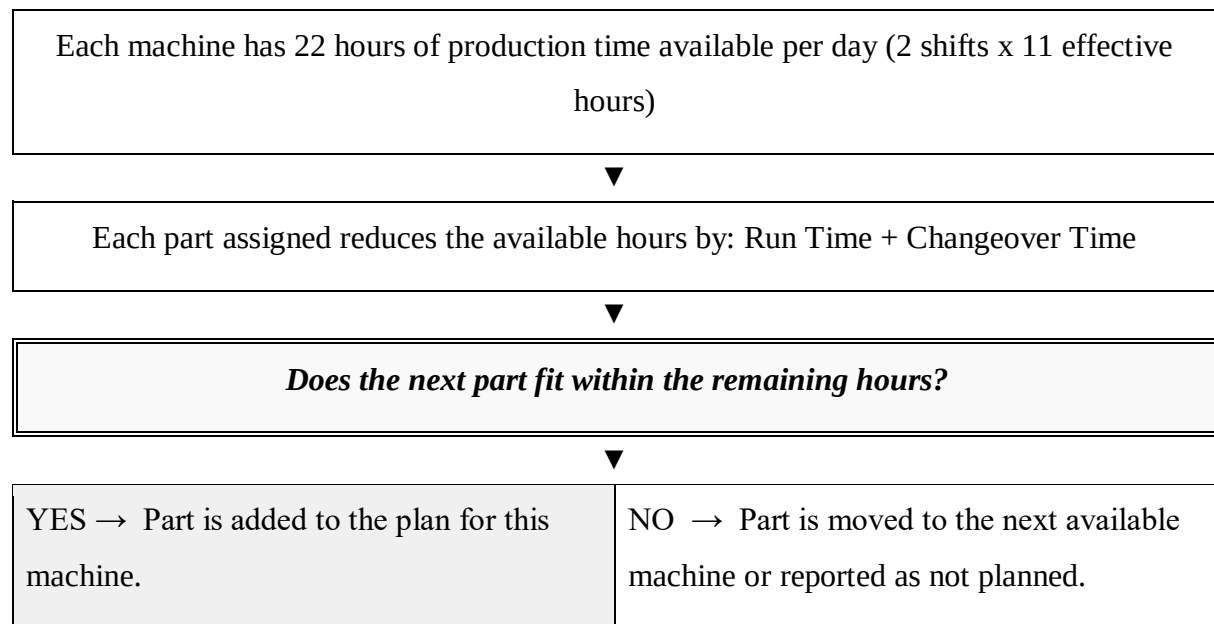


Figure 1: Decision flow for the 22-hour shift rule.

If It Still Fails

If a part cannot fit on any compatible machine that day, it is placed on a report called VT_Not_Planned. This report shows the exact reason the part could not be scheduled and the precise number of free hours remaining on each compatible machine. The planner can then decide whether to adjust quantities of lower-priority parts to create space, or defer the part to the following day.

5. CONSTRAINT 4: SPECIALISED MACHINES MUST BE PROTECTED

The Problem

Some machines on the factory floor can only run two or three specific parts. These are called specialised machines. If a general-purpose part is assigned to a specialised machine and uses up its available hours, the parts that only that machine can produce have nowhere to go for the entire day. They cannot be made, even though the right machine exists. This creates avoidable stock-outs for those specific parts.

Our Solution

Smart APS reads a data sheet called VT_Machine_Part_Count, which records how many parts each machine is capable of running. When choosing which machine to assign a part to, the system always prefers the machine with the fewest compatible parts. A machine that runs only two parts is rated higher than a machine that runs fifteen parts. This ensures that specialised machines are claimed by their dedicated parts before any general part can occupy them.

Figure 2 shows an example of how the machine ranking works.

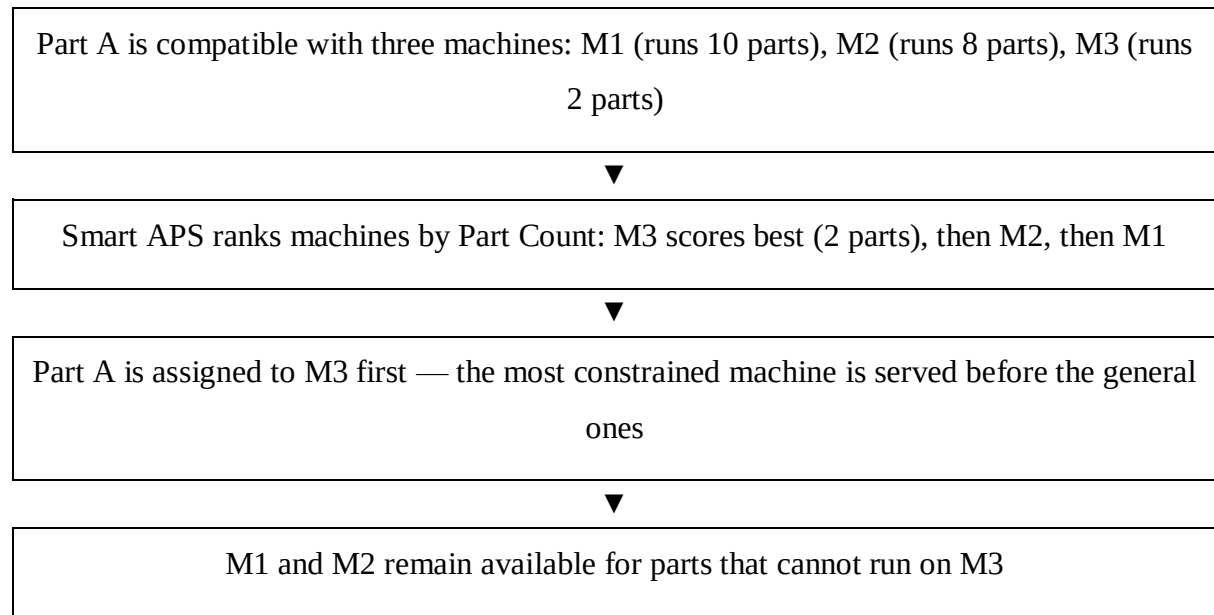


Figure 2: Specialised machine protection through Part Count ranking.

If It Still Fails

If even the specialised machine is fully occupied and a dedicated part cannot be planned, it appears in the VT_Not_Planned report with the reason clearly stated. The planner can then review the specialised machine's schedule and free up hours if necessary.

6. CONSTRAINT 5: CONTINUE THE SAME PART TO AVOID UNNECESSARY CHANGEOVERS

The Problem

At the end of every day, each machine is left with a mould loaded inside it — whatever part it was running last. The next day, two situations can arise. First, if the same part is planned again on that machine, no changeover is needed and production starts immediately. Second, if a different part must be run on that machine today, the old mould must be removed and a new one fitted before production can begin, consuming approximately 40 minutes.

The more serious problem arises when many machines across the floor all need a changeover at the start of the shift. This happens when the parts that ran yesterday are not part of today's plan on several machines simultaneously. Since there is only one tool-change team for all 30 machines, they cannot attend to multiple machines at the same time. Many machines end up waiting, and the shift loses significant production time right from the start.

Our Solution

Smart APS handles this in two stages. First, at the end of each day it saves the last part run on every machine to a file called `machine_state.json`. The next day, when assigning parts to machines, the system gives strong preference to placing each part on the machine it was already running on. This directly reduces the number of changeovers needed at the start of the shift.

Second, for machines where a changeover is unavoidable, the system does not treat all these changeovers as happening simultaneously. The tool-changer serial queue described in Constraint 7 takes all shift-start and mid-shift changeovers and sequences them into one non-overlapping queue. While a machine waits for the tool-change team to arrive, it does not sit idle — Smart APS automatically extends the run of the previous part on that machine to fill the entire waiting period. The machine keeps producing right up until the moment the changeover begins. The planned quantity for the new part is then calculated against the actual run time remaining after the changeover. Table 1 below summarises the three possible situations.

	Case 1: Same Part	Case 2: Different Part, Changeover Early in Queue	Case 3: Different Part, Changeover Late in Queue
Mould status	Already loaded	Needs to be changed	Needs to be changed
While waiting	No wait	Previous part's run extended to fill the wait	Previous part's run extended to fill the wait
New part start time	00:00 immediately	Shortly after shift start	Later in the shift
Effective run time (new)	Full 22 hours	22 hours minus wait	22 hours minus longer wait
Planned qty (new part)	Based on 22 hours	Reduced to match actual run time	Further reduced to match actual run time

Table 1: The three possible situations at shift start and their effect on planned quantities.

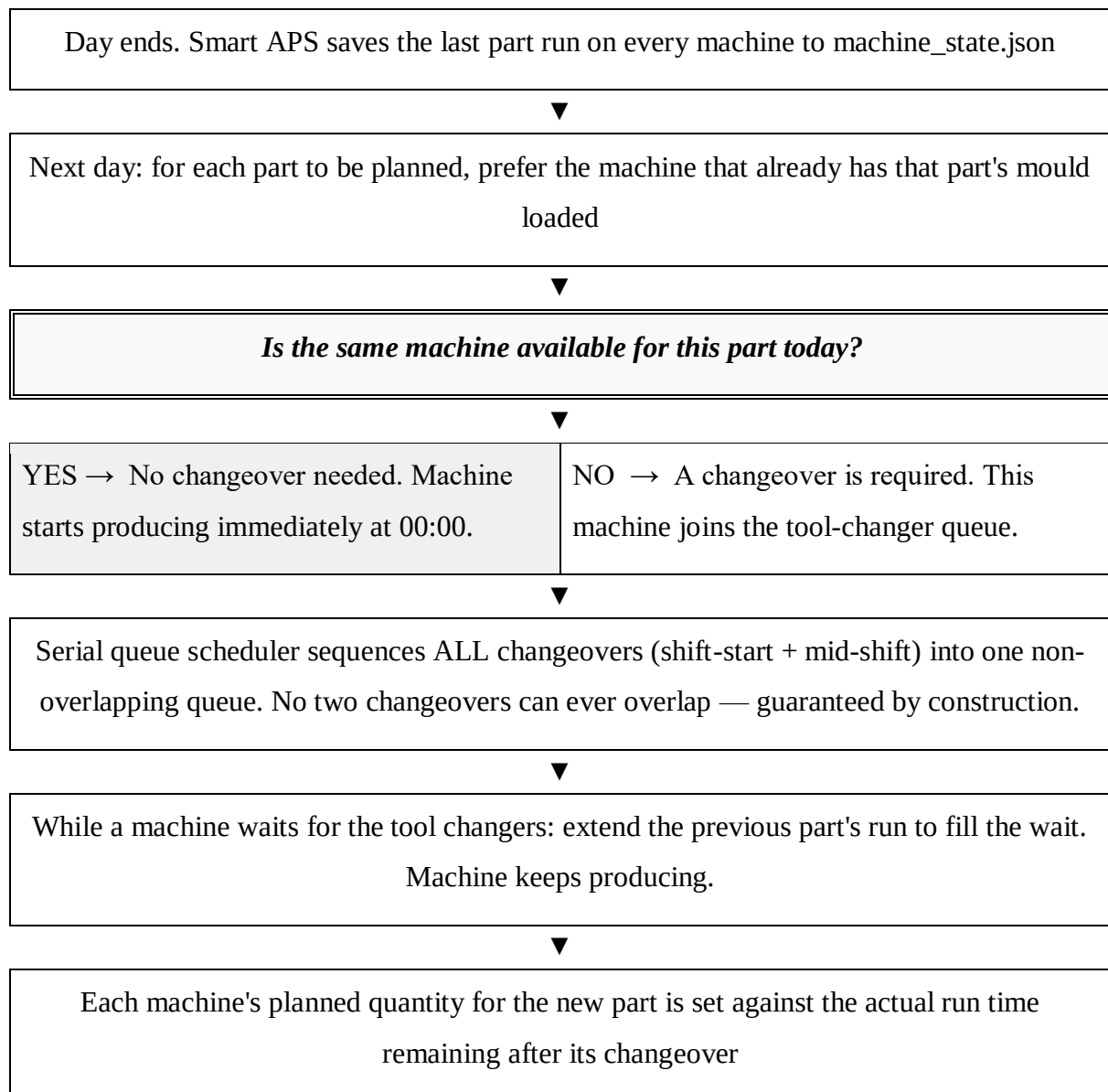


Figure 2: How Smart APS minimises changeovers, fills wait time with production, and adjusts planned quantities.

If It Still Fails

The previous part always runs during the wait, so no production time is truly lost — the machine is producing something useful right up to the moment of changeover. However, the new part ends up with fewer run hours than a machine that needed no changeover. If the reduced quantity is not enough to meet the daily requirement for the new part, Smart APS will look for another compatible machine to cover the shortfall using the minimum-tools logic described in Constraint 6. If no additional machine is available, the gap is reported in VT_Not_Planned.

7. CONSTRAINT 6: USE THE MINIMUM NUMBER OF MOULDS

The Problem

Each part has a physical mould (tool) that must be fitted to a machine to produce it. Some parts have more than one mould available. Using too many moulds for a single part simultaneously ties up multiple machines unnecessarily, reducing the capacity available for other parts. The goal is always to produce the required quantity using as few machines as possible.

Our Solution

Smart APS always attempts to produce the full daily requirement for a part on a single machine first. Only if that machine does not have enough hours left in the shift does the system bring in a second machine. When a second machine is used, it only runs for the hours needed to cover the shortfall from the first machine, not the full requirement again. A third machine is only used if both the first and second together are still insufficient. Figure 3 shows this decision process.

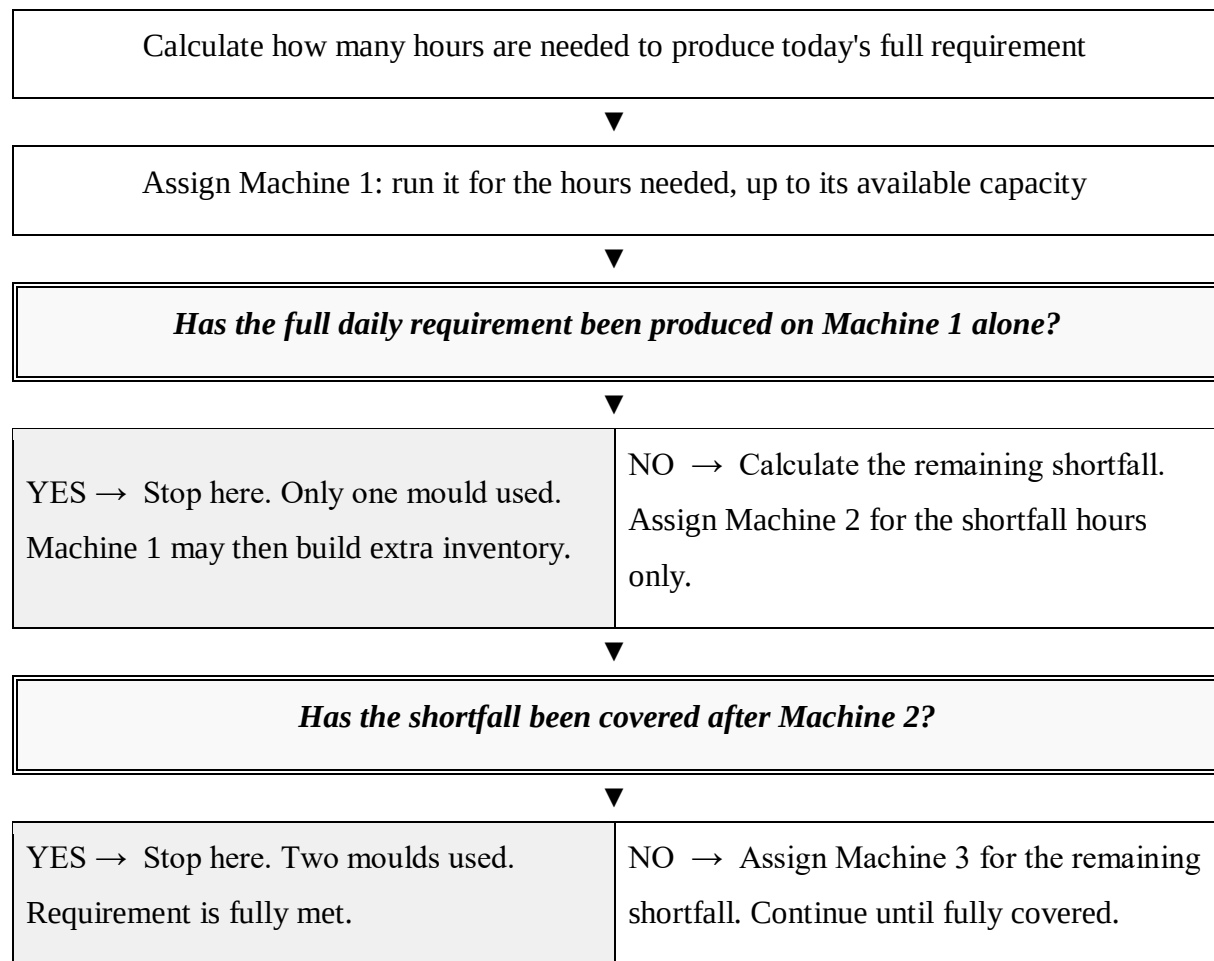


Figure 3: Minimum tools logic — each additional machine covers only the remaining shortfall.

If It Still Fails

If the full daily requirement cannot be met even with all available moulds deployed, the system reports the gap in the VT_Production_vs_Indent sheet, showing exactly how much was produced versus how much was required. The planner can then decide whether the shortfall can be covered from existing inventory.

8. CONSTRAINT 7: ONE TOOL CHANGER FOR ALL 30 MACHINES

The Problem

There is a single shared tool-change team that performs all mould changeovers across the entire VT machine floor. They can only work on one machine at a time. If the plan requires several machines to undergo a changeover at the same moment, the tool changers can only attend to one of them. The others must wait, sitting completely idle during that time. This idle time is wasted production capacity.

Our Solution

After the main plan is built, Smart APS collects every changeover event across all machines and sequences them using a strict serial queue. The queue works as follows: changeovers are sorted by the time at which each machine naturally needs them. They are then processed one by one. Each changeover can only begin after the previous one is fully complete, plus a mandatory twenty-minute buffer after the previous changeover finishes. This structure makes it physically impossible for two changeovers to overlap — the guarantee comes from the mechanism itself, not from an optimisation solver. Furthermore, when a machine must wait because the tool changers are busy elsewhere, the system automatically extends the run of the current part on that machine to fill the entire waiting period. The machine continues producing instead of standing idle.

Figure 4 shows how this works.

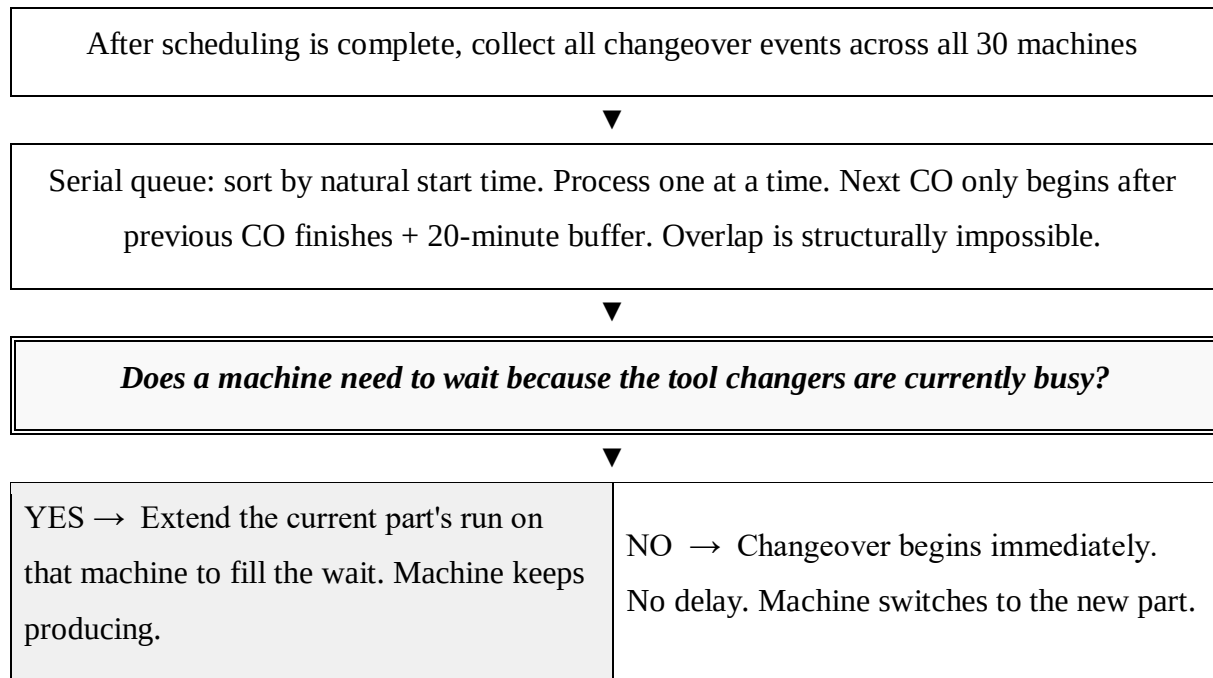


Figure 4: Tool-changer sequencing and machine wait resolution.

If It Still Fails

The serial queue scheduler processes changeovers one at a time, in order of when each machine needs them. Because the next changeover can only begin after the previous one is fully complete, two changeovers cannot physically overlap — this is guaranteed by the structure of the algorithm itself, not by a solver or a time limit. The plan is always produced in milliseconds regardless of how many changeovers are queued, and the no-overlap rule cannot be broken under any circumstances.

9. CONSTRAINT 8: MINIMUM RUN BLOCK OF FOUR HOURS

The Problem

Setting up a mould on an injection moulding machine (called a changeover) takes approximately 40 minutes. If a part is only planned to run for one or two hours after this setup, the changeover time represents a very large proportion of the total time used, and very few parts are produced. This makes poor use of the machine and the tool-change team's time.

Our Solution

Smart APS will only assign a part to a machine if, after subtracting the changeover time, at least four hours remain for actual production. If only three hours are available after the changeover, the system does not assign that part to that machine. It moves on to find another machine that has more capacity. Figure 5 illustrates this check.

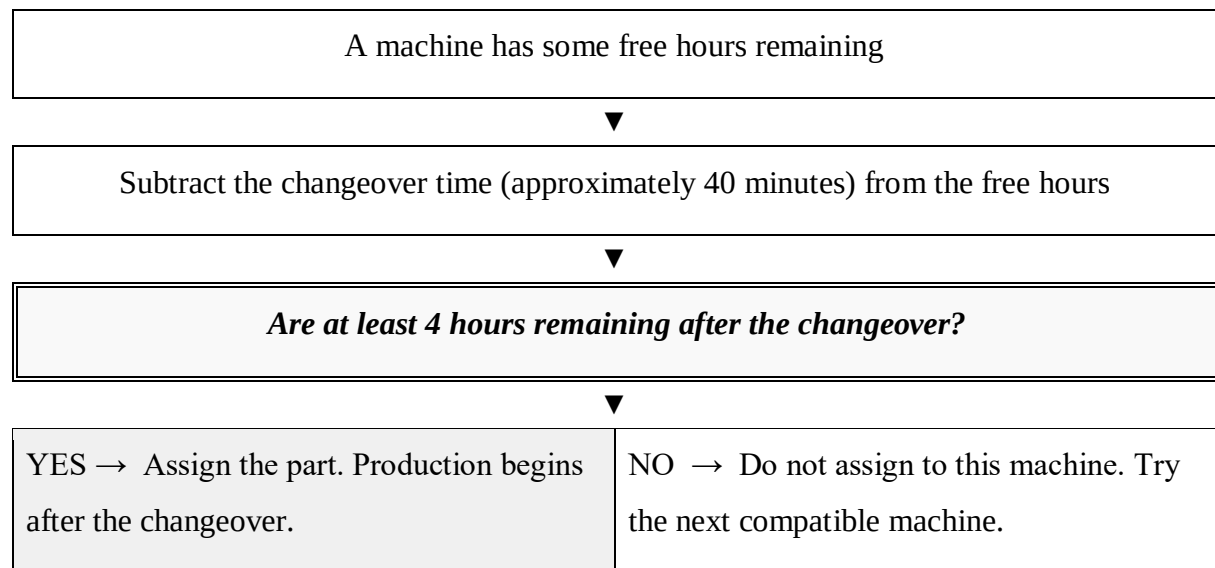


Figure 5: Minimum run block check.

If It Still Fails

If no machine has four or more effective hours available for a particular part, that part is added to the VT_Not_Planned report. The planner can review it and, if necessary, reduce the planned quantity of another part on a suitable machine to free up enough time.

10. CONSTRAINT 9: NO MACHINE SHOULD SIT IDLE

The Problem

After all the urgent parts have been planned, some machines will have hours remaining in their shift. If nothing is done with this remaining capacity, those hours are simply wasted. Across 30 machines, even a small amount of idle time per machine per day adds up to a significant loss of production capacity over time.

Our Solution

After the main planning pass is complete, Smart APS runs a second pass called the Utilisation Enforcer. It checks every machine for remaining free hours and attempts to fill them in two steps, as shown in Figure 6.

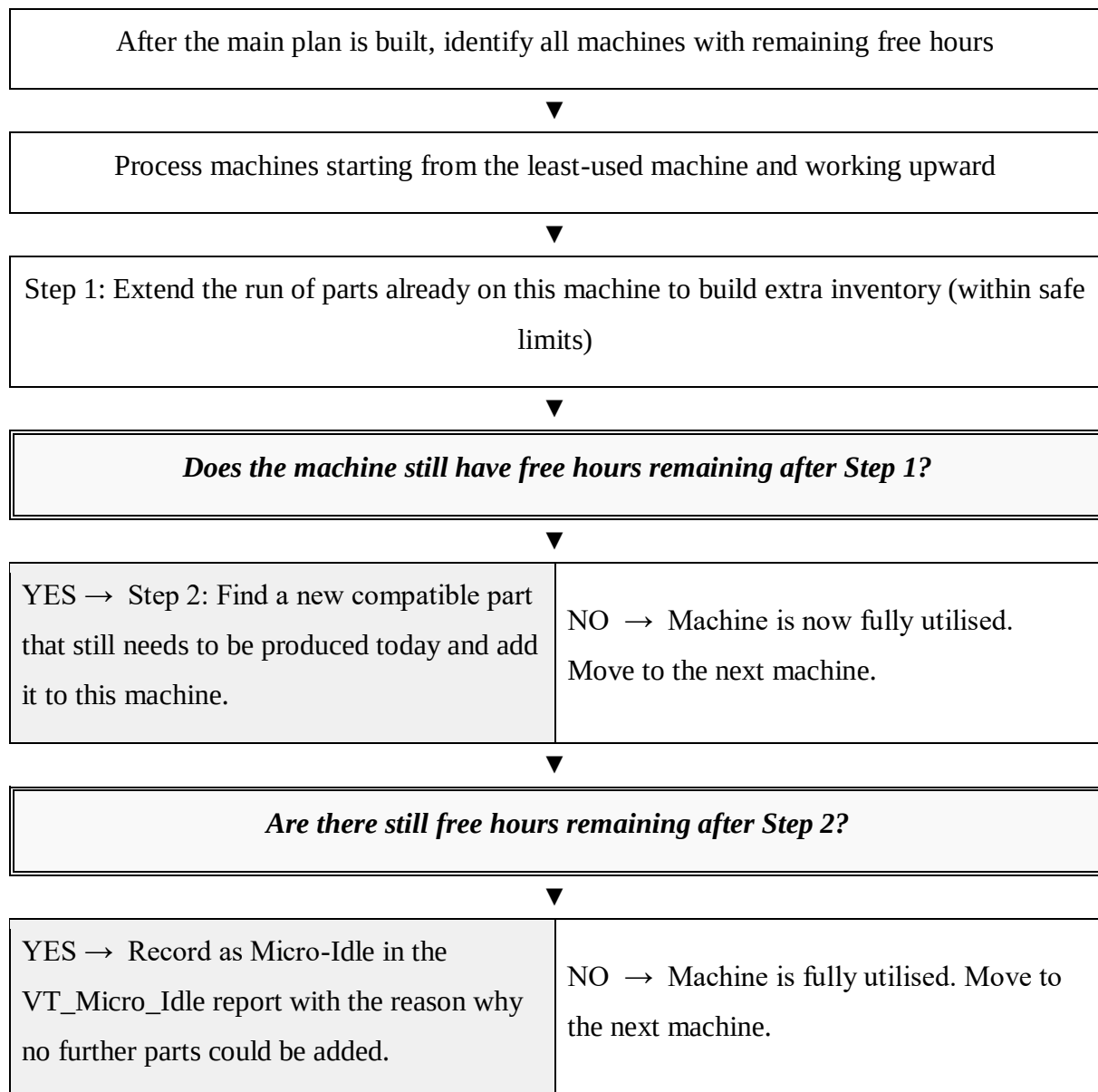


Figure 6: The Utilisation Enforcer — two-step process for filling idle machine time.

If It Still Fails

If a machine still has idle time after both steps, it means all compatible parts have either already been planned or have reached their maximum allowable production quantity for the day. This residual idle time is logged in the VT_Micro_Idle sheet with a full explanation, giving management complete visibility of why the machine could not be filled.

11. CONSTRAINT 10: GROUP SAME-COLOUR PARTS ON THE SAME MACHINE

The Problem

Each part is moulded in a specific colour. When a machine switches from one part to another part of a different colour, the mould must be purged to remove residual colour from the previous material. This purging process adds an extra ten minutes to the changeover time. If colour changes happen frequently across the shift, the cumulative purging time can consume a significant portion of the day's available hours.

Our Solution

Smart APS reads the colour of every part from the VT sheet. When assigning a part to a machine, the system gives preference to the machine whose last-run part shares the same colour. This way, the colour purge is avoided as often as possible. The ranking of machines follows this preference order, from most preferred to least preferred.

Situation	Changeover Time	Why
Same part as last run	Zero minutes (no changeover at all)	Mould is already loaded and ready. Production starts immediately.
Different part, same colour	Normal changeover time only (no purge)	Mould changes but colour is the same. No residual to clear.
Different part, different colour	Normal changeover time plus ten minutes for purging	Old colour must be purged from the barrel before the new part can run.

Table 5: How colour affects the total changeover time.

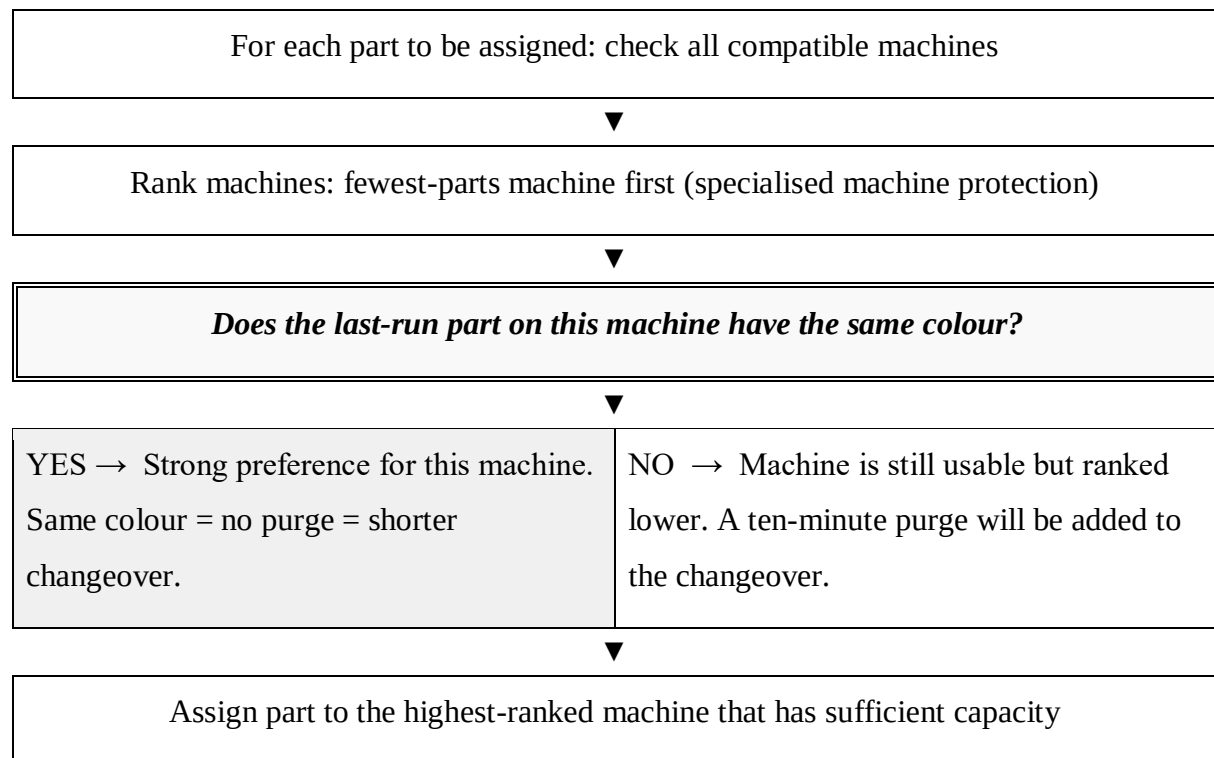


Figure 7: Machine ranking with colour preference built in.

If It Still Fails

If no compatible machine has the same colour as the new part, the system assigns the best available machine regardless of colour and adds the ten-minute purge to the changeover time. This purge time is included in the twenty-two-hour daily calculation so the machine is never over-committed. The quantity-based changeover stagger (Constraint 7) also accounts for the longer changeover duration when colours differ, ensuring the tool-change team is never called on two machines at the same time.

11. SUMMARY OF ALL CONSTRAINTS

Table 4 below provides a concise reference summary of all ten constraints, their solutions, and their fallback actions.

No.	Constraint	Solution	Fallback
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1	Quantities planned depend on inventory health	Daily scenario classification (S0 to S3) sets the quantity logic for all parts.	Scenario and reason printed on every output sheet for planner awareness.
2	Most urgent parts must be planned first	Priority score (0 to 100) computed per part. Parts assigned in score order.	Highest-priority unplanned parts shown at the top of VT_Not_Planned.
3	Machine cannot exceed 22 hours per day across both shifts	Hard ceiling: available hours tracked at all times across both shifts.	Part logged in VT_Not_Planned with free-hours detail.
4	Specialised machines must not be taken by general parts	Part Count ranking: machine with fewest compatible parts is always preferred first.	If specialised machine is full, appears in VT_Not_Planned for planner action.
5	Continuing the same part avoids a changeover	Machine state saved daily. Same-part continuation strongly preferred in machine ranking.	If unavailable, changeover is charged and sequenced by the tool-changer scheduler.
6	Use the minimum number of moulds	Try one machine first. Add a second only for the remaining shortfall. Repeat if necessary.	Gap between required and produced shown in VT_Production_vs_Indent report.
7	One tool changer for all 30 machines	Serial queue sequences all changeovers so no two overlap. Machine waits are filled with extended production. Overlap is structurally impossible.	No fallback needed. Serial queue guarantees no overlap by construction, not by optimisation.

8	Minimum four-hour run block per part	Part only assigned if four or more effective hours remain after changeover.	Next compatible machine is tried. If none, part goes to VT_Not_Planned.
9	No machine should sit idle	Utilisation Enforcer extends existing runs, then adds new parts to fill remaining hours.	Residual idle time logged in VT_Micro_Idle with the reason.
10	Group same-colour parts on the same machine	Machine ranking prefers same colour as last-run part. Different colour adds 10-min purge to changeover.	Colour mismatch accepted if no same-colour machine available. Purge time included in 22-hour budget.

Table 4: Summary of all ten constraints, solutions, and fallback actions.

13. APPROVAL AND SIGN-OFF

This document has been prepared to ensure that all stakeholders, irrespective of their technical background, have a complete understanding of the rules governing the Smart APS system before it is deployed for daily production use.

By signing below, the authorised personnel confirm that they have read and understood the constraints described in this document and approve the Smart APS V9 system for use as the primary production scheduling tool for the Injection Moulding Division.

Prepared By (Intern): Tushar:

Name: _____

Date: _____

Guided By: Mr. Amrit Lal:

Name: _____

Date: _____

Approved By: Plant Manager / VP Operations:

Name: _____

Date: _____

Document Version History

Version	Date	Prepared By	Summary of Changes
V1.0	March 2026	Tushar, under guidance of Mr. Amrit Lal	Initial version (V8) — ten constraints, colour constraint, serial queue tool changer
V2.0	March 2026	Tushar, under guidance of Mr. Amrit Lal	V9 upgrade: five-day inventory target (SAFETY_DAYS=3, TARGET_DAYS=5), displacement pass

			for zero-inventory Runners/Repeaters, serial queue tool-changer scheduler, colour constraint fully implemented with 10-min purge penalty and colour grouping in all output sheets, VT_Inventory_Target new output sheet.